

ACF & PACF

Partial correlation

For $X, Y \in \mathbb{R}$ r.v.s.

we can compute $\text{corr}(X, Y) = \frac{\text{Covariance}(X, Y)}{\sqrt{\text{var}(X) \text{var}(Y)}}$

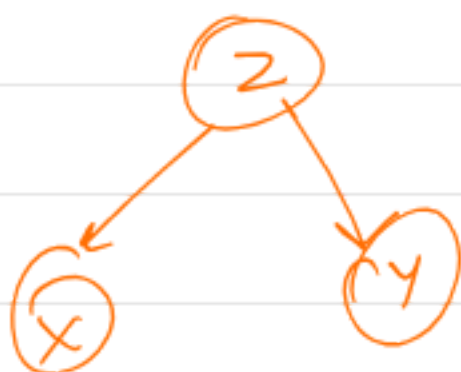


$X = \text{price of your House}$
 $Y = \text{price of your Car}$

↓
 Not correlated directly



Hidden $Z = \text{Income}$



Conditional on Z
 X & Y are uncorrelated

To understand such relations between x, y, z , we use Regression,

$$\begin{aligned}\text{Partial Corr.} &= \text{corr}(x, y | z) \\ &= \text{corr}(x - \hat{x}, y - \hat{y})\end{aligned}$$

where $\hat{x}$ & $\hat{y}$ are fitted values for Regressing x & y on z resp.

$$\begin{aligned}\hat{x}_i &= \hat{\alpha}_0 + \hat{\alpha}_1 z_i \\ \hat{y}_i &= \hat{\beta}_0 + \hat{\beta}_1 z_i\end{aligned} \quad \begin{array}{l} \swarrow \text{Two simple Reg.} \\ \swarrow \text{model} \end{array}$$

Setting: If $\text{corr}(x, y | z) \approx 0$
then x, y are conditionally
uncorrelated.

If x, y, z are Normal, then we have
cond. Independence.

↳ can have any relation

$$\text{corr}(x - \hat{x}, y - \hat{y}) = \text{corr}(x, y | z)$$

$$= \frac{E[(x - EX)(y - EY) | z]}{\sqrt{E[(x - EX)^2 | z] E[(y - EY)^2 | z]}}$$

Partial Auto correlation

Let X_t be stationary $\rho \rightarrow$ autocorr.

Partial autocorr.

$$\psi(h) = \begin{cases} \rho_x(1) & : \text{if } h=1 \\ \text{corr}(X_{t+h} - \hat{X}_{t+h}, X_t - \hat{X}_t) & : \text{if } h>1 \end{cases}$$

where $\hat{X}_{t+h}$ & $\hat{X}_t$ comes from linear regression on $X_{t+h-1} \dots X_{t+1}$

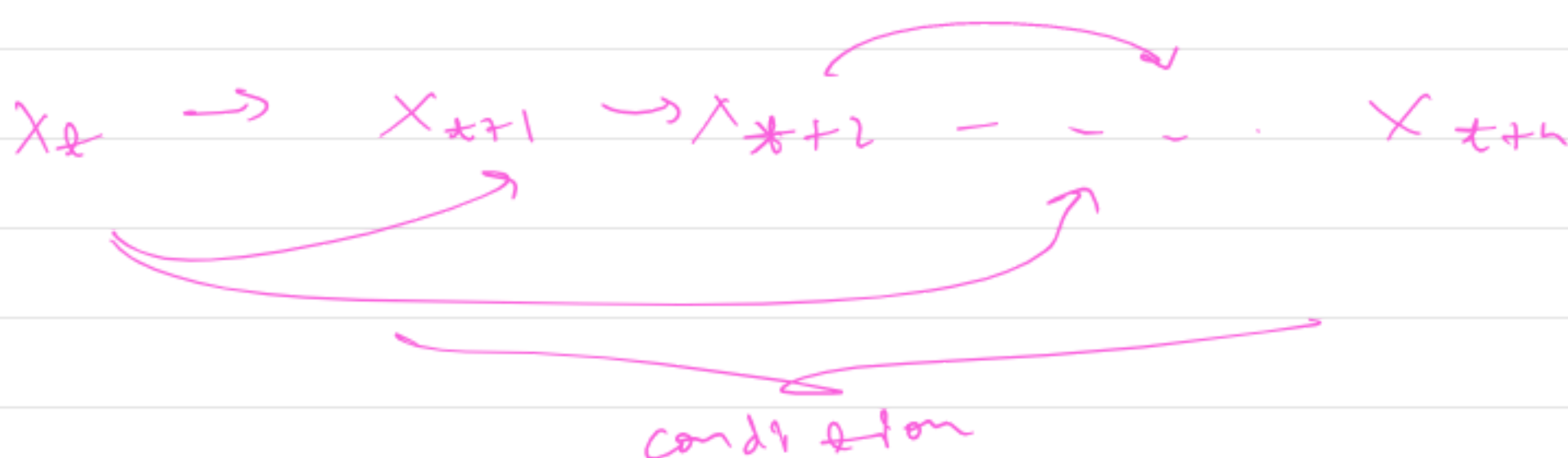
i.e. $\hat{X}_{t+h} = \beta_1 X_{t+h-1} + \dots + \beta_{h-1} X_{t+1}$

$$\hat{X}_t = \beta_1 X_{t+1} + \dots + \beta_{h-1} X_{t+h-1}$$

Same β 's due to sym. of autocorr.

Also if X_t is a Gaussian Process

then $\text{corr}(\hat{X}_{t+h}, \hat{X}_t \mid X_{t+h-1} \dots X_{t+1})$



Basically, like Markov chains, where X_t & X_{t+h} can't be connected unless $h=1$, we try to check if X_t & X_{t+h} are related if remove the middle points.

ACF and PACF for AR(p) and MA(q)

1) ACF for AR(p)

Consider the AR(1) process

$$X_t = \phi X_{t-1} + \omega_t \quad \text{where } |\phi| < 1$$

$\therefore$ Stationary & Mean 0

Note: (Multiply by X_{t-h})

$$X_t X_{t-h} = \phi X_{t-1} X_{t-h} + \omega_t X_{t-h}$$

$$E[X_t X_{t-h}] = \phi E[X_{t-1} X_{t-h}] + E[\omega_t X_{t-h}]$$

$$K_x(h) = \phi K_x(h-1) + 0$$

$\because \omega_t$ & X_{t-h} are uncorrelated

First order difference eqⁿ

$$f(h) = \phi f(h-1)$$

Charac. Poly = $1 - \phi z$ with root $\phi^{-1} = z_1$

We solve it to get

$$f(h) = c z_1^{-h} \quad \text{or} \quad \sigma_h = c \phi^h \quad \text{with} \\ c \text{ any constant}$$

Check

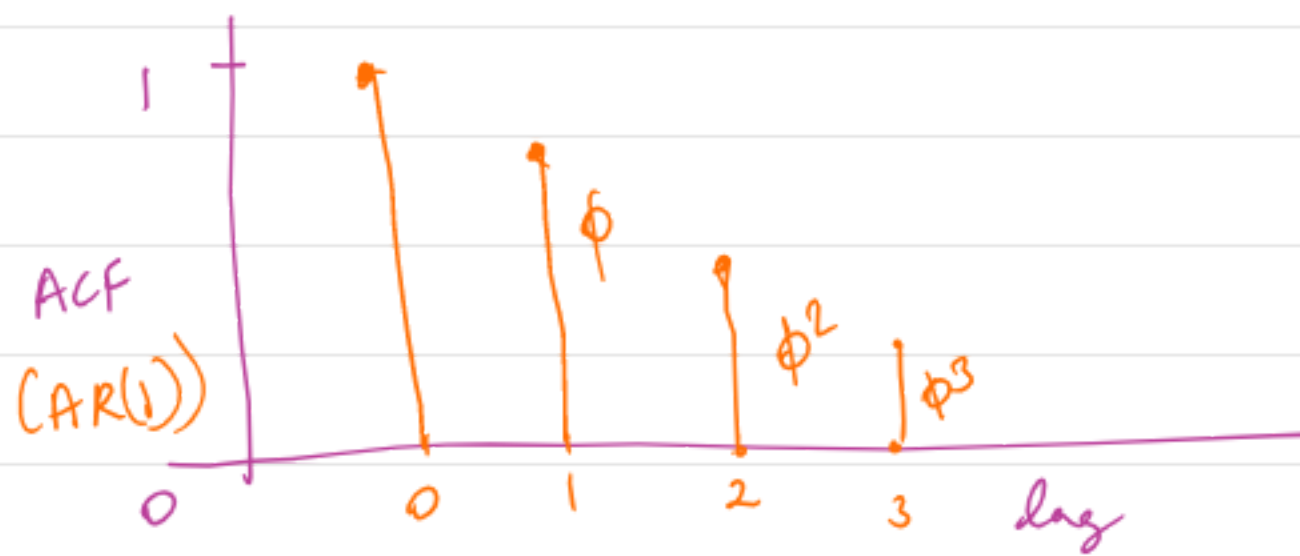
$$c \phi^h = \phi \times c \phi^{h-1}$$

- "c" is the initial condition $f(0) = c$
- for autocorr., $K_x(0) = \text{var}(x_t)$

so, $f_x(h)$ has the same diff. eqⁿ but with $c=1$.

"So $\rho_x(h)$ has the same diff. eqn, but with $c=1$ " — this connects to the fact above. $\rho_x(h)$ (autoCORRELATION, normalized version) is just $K_x(h)$ divided by $K_x(0)$ (i.e., divided by the variance) — that's literally how you turn covariance into correlation, same as always (correlation = covariance / product of standard deviations). So $\rho_x(h)$ satisfies the exact same underlying differential equation as $K_x(h)$ — same decay pattern, same dynamics — except when you plug in the "starting value" for ρ at $h=0$, you get $\rho_x(0) = K_x(0)/K_x(0) = 1$ (since anything divided by itself is 1). That matches what you already intuitively know: a series' correlation with itself at zero lag is always exactly 1, no exceptions.

$$\therefore \rho_x(h) = \phi$$



AR(2)

$$E [X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \omega_t \quad \times \quad X_{t-h}]$$

$$\Rightarrow E[X_t X_{t-h}] = \phi_1 E[X_{t-1} X_{t-h}] + \phi_2 E[X_{t-2} X_{t-h}] + \underline{\underline{0}}$$

needs to be stat.

$$K_x(h) = \phi_1 K_x(h-1) + \phi_2 K_x(h-2)$$

$$\text{Charact. Poly} \rightarrow 1 - \phi_1 z - \phi_2 z^2$$

with roots z_1 and $z_2 \in \mathbb{C}$

Assume that $|z_1| > 1$ & $|z_2| > 1$

so that X_t is causal & stationary

Solving a second order diff. eqⁿ

$$f(h) = \phi_1 f(h-1) + \phi_2 f(h-2)$$

⇒ There are 3 possible solutions based on the roots:

1) $z_1 \neq z_2$ and in $\mathbb{R}$ then,

$$f_x(h) = c_1 z_1^{-h} + c_2 z_2^{-h}$$

Geometric Decay
in ACF
like AR(1)

where $c_1 + c_2 = 1$ a initial condition
 $h=0$

2) $z_1 = z_2$ (Must be $\mathbb{R}$)

$$f_x(h) = z_1^{-h} (c_1 + c_2 h)$$

↖ G. Decay
↘ Linear

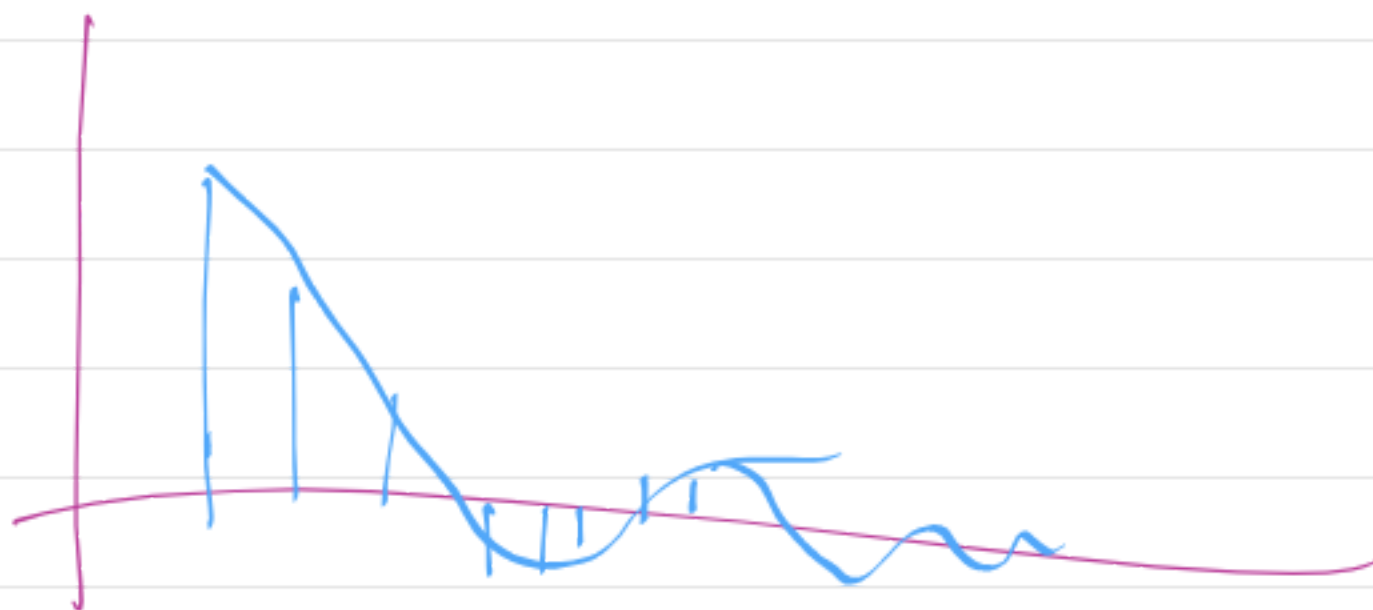
3) $z_1 = \bar{z}_2 \in \mathbb{C}$ (complex conjugate roots)

$$f_x(h) = c_1 z_1^{-h} + c_2 \bar{z}_1^{-h}$$

$$= |c_1| |z_1|^{-h} \left(e^{-i\theta} e^{-i\theta h} + e^{i\theta} e^{i\theta h} \right)$$

$$= 2 |c_1| |z_1|^{-h} \underbrace{\cos(\theta h + \theta)}_{\text{osc.}}$$

Decay



We could consider $AR(p)$ with

$$f_x(h) = z_1^{-h} f_1(h) + \dots + z_n^{-h} f_n(h)$$

where root z_i has multiplicity m_i s.t.
 $\sum_{i=1}^n m_i = p$ and $f_i(h)$ is an m_i th degree poly in h .

2) ACF for $MA(q)$

Plain English first:

$MA(q)$ = Moving Average model of order q . It says: today's value depends on today's random noise PLUS a weighted combination of the last q noise terms (not past values of the series itself — that's AR's job — but past shocks/errors).

Formula, in the same style as $AR(p)$ you already saw:

$$X_t = e_t + \theta_1 e_{t-1} + \theta_2 e_{t-2} + \dots + \theta_q e_{t-q}$$

where e_t, e_{t-1} , etc. are random noise terms (typically white noise — random, mean 0, no autocorrelation among themselves).

Contrast with $AR(p)$, since you already understand that one:

- **AR(p)**: today's value depends on past values of the series itself ($X_{t-1}, X_{t-2}, \dots$) — "I remember my own past levels."
- **MA(q)**: today's value depends on past random shocks/errors ($e_{t-1}, e_{t-2}, \dots$) — "I remember recent surprises/noise, and each surprise has a lingering effect for q days before fading out completely."

assume $\sigma^2 = 1$

$$\text{Let } X_t = \sum_{j=0}^a \theta_j w_{t-j} \quad \text{with } \theta_0 = 1$$

$$\text{Then, } K_X(h) = \sigma^2 \sum_{j=0}^{a-h} \theta_j \theta_{j+h} \quad \text{for } h=0 \dots a \\ = 0 \quad \text{otherwise}$$

$$\therefore K_X(0) = 1 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \dots$$

$$\therefore \rho_X(h) = \frac{\sum_{j=0}^{a-h} \theta_j \theta_{j+h}}{\sum_{j=0}^a \theta_j^2} \quad \text{for } h \leq a \\ 0 \quad \text{for } h > a$$

Hence, the ACF vanishes when $h > a$,

$\therefore$ we can use it to determine the order of the MA process

$$\text{For MA}(1), \text{ we have } \rho_X(1) = \frac{\theta_1}{1 + \theta_1^2}$$

Partial Autocorrelation (PACF)

X_t stationary, lag h

$$\Psi_x(h) = \begin{cases} \rho_x(1) & , h=1 \\ \text{corr}(X_{t+h} - \hat{X}_{t+h}, X_t - \hat{X}_t) & h>1 \end{cases}$$

where $\hat{X}_t$ and $\hat{X}_{t+h}$ come from regressing X_t and X_{t+h} (resp) on the intern. points $X_{t+1} \dots X_{t+h-1}$.

⇒ PACF for AR(p)

Consider the simple AR(1)

$$X_t = \phi X_{t-1} + w_t, \text{ with } |\phi| < 1$$

first compute

$$\text{corr}(X_t, X_{t-2}) =$$

$$\underbrace{\phi w_{t-1} + w_t}_0$$

$$\begin{aligned} &= \text{corr}(\phi X_{t-1} + w_t, X_{t-2}) \\ &= \text{corr}(\phi^2 X_{t-2} + \phi w_{t-1} + w_t, X_{t-2}) \\ &= \text{corr}(\phi^2 X_{t-2}, X_{t-2}) \\ &= \phi^2 \text{corr}(X_{t-2}, X_{t-2}) \\ &= \phi^2 \end{aligned}$$

X_{t-2} does not depend on future noise



For the Partial Autocorrelation, we have

$$\begin{aligned} & \text{corr}(X_t - \phi X_{t-1}, X_{t+2} - \phi X_{t+1}) \\ &= \text{corr}(\underbrace{\omega_t, X_{t+2} - \phi X_{t+1}}_{\text{Not correlated}}) = 0 \end{aligned}$$

$$\therefore \psi(x) = \begin{cases} \phi, & h=1 \\ 0, & h>1 \end{cases}$$

Why did we choose $\hat{X}_t$ to be ϕX_{t-1} ?

To Set this, we compute a least square est.

$$\hat{X}_t \text{ for } X_t, \text{ eg } \hat{X}_t = \hat{\beta} X_{t-1}$$

Choose $\hat{\beta}$ to minimize

$$\begin{aligned} & E \left([X_t - \beta X_{t-1}]^2 \right) \quad \text{Assuming mean zero} \\ &= E X_t^2 - 2\beta E [X_t X_{t-1}] + \beta^2 E [X_{t-1}^2] \\ &= K_x(0) - 2\beta K_x(1) + \beta^2 K_x(0) \\ &= (1 + \beta^2) K_x(0) - 2\beta K_x(1) \end{aligned}$$

Critical Point by Diff.

$$\frac{d \{ \} }{d\beta} = 2\beta K_x(0) - 2K_x(1) = 0$$

$$\Rightarrow \hat{\beta} = \frac{K_x(1)}{K_x(0)} = \rho_x(1) = \phi$$

$$\Rightarrow X_t = \phi X_{t-1} + \omega_t$$

$$E[(X_{t+1} - \beta X_{t-1})^2] =$$

$$E[X_{t+1}^2] - 2\beta E[X_{t+1}X_{t-1}] + \beta^2 E[X_{t-1}^2]$$

$$= K_x(0) - 2\beta K_x(1) + \beta^2 K_x(0)$$

$$= (1+\beta^2) K_x(0) - 2\beta K_x(1)$$

$\therefore$ Same $\hat{\beta} = \phi$ in this case

In AR(p),

$$X_t = \omega_t + \sum_{i=1}^p \phi_i X_{t-i}$$

For lags $> p$, we assume that

$$\text{LSE is } \hat{X}_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + 0 X_{t-p-1} + \dots$$

$$\hat{X}_{t-h} = \phi_1 X_{t-h+1} + \dots + \phi_p \hat{X}_{t-h+p}$$

$$\text{corr}(X_t - \hat{X}_t, X_{t-h} - \hat{X}_{t-h})$$

$$= \text{corr}(w_t, X_{t-h} - \hat{X}_{t-h}) = 0$$

the PACF = 0 for AR(p) if $h > p$

On the other hand, computing the PACF for lags $\leq p$ is hard in general.

PACF for $MA(q)$

Once again consider $MA(1)$

$$X_t = w_t + \theta w_{t-1} \quad \text{with } |\theta| < 1$$

so it's Invertible

$$\therefore \text{ We can write } X_t = w_t + \sum_{i=1}^{\infty} \theta^i X_{t-i}$$

$$\therefore \text{ We get that for } \hat{X}_t = \hat{\beta} X_{t-1} \quad \hat{\beta} = \frac{\theta}{1+\theta^2}$$

$$\hat{\beta} = \frac{K_X(1)}{K_X(0)}$$

$$\text{Cov} \left(X_t - \frac{\theta}{1+\theta^2} X_{t-1}, X_{t-2} - \frac{\theta}{1+\theta^2} X_{t-1} \right)$$

$$= \text{Cov}(X_t, X_{t-2}) - \frac{\theta}{1+\theta^2} \text{Cov}(X_t, X_{t-1}) \\ - \frac{\theta}{1+\theta^2} \text{Cov}(X_{t-1}, X_{t-2}) + \frac{\theta^2}{(1+\theta^2)^2} \text{Cov}(X_{t-1}, X_{t-1})$$

$$= K_X(2) - \frac{2\theta}{1+\theta^2} K_X(1) + \frac{\theta^2}{(1+\theta^2)^2} K_X(0)$$

$$= 0 - \frac{2\theta^2}{1+\theta^2} + \frac{\theta^2}{(1+\theta^2)^2} = \frac{-\theta^2}{1+\theta^2}$$

$$\bullet \text{Var} \left(X_t - \frac{\theta}{1+\theta^2} X_{t-1} \right) = K_X(0) \left[1 + \frac{\theta^2}{(1+\theta^2)^2} \right] - K_X(1) \frac{2\theta}{1+\theta^2}$$

$$= 1 + \theta^2 + \frac{\theta^2}{1+\theta^2} + \frac{2\theta^2}{1+\theta^2} = \frac{1+\theta^2+\theta^4}{1+\theta^2}$$

$$\therefore \text{PACF } \psi(1) = \frac{\left(\frac{-\theta^2}{1+\theta^2} \right)}{\frac{1+\theta^2+\theta^4}{1+\theta^2}}$$

$$= \boxed{\frac{-\theta^2}{1+\theta^2+\theta^4}}$$

↙ PACF
MA(1)

Point $\Rightarrow$ PACF for MA(1) decreases $\rightarrow 0$ in
Magnitude but doesn't vanish.

AR(p)

MA(q)

ACF

↘ Geometric

= 0 for $h > q$

PACF

= 0 for
 $h > p$

↘ Geometric